Comptia N+

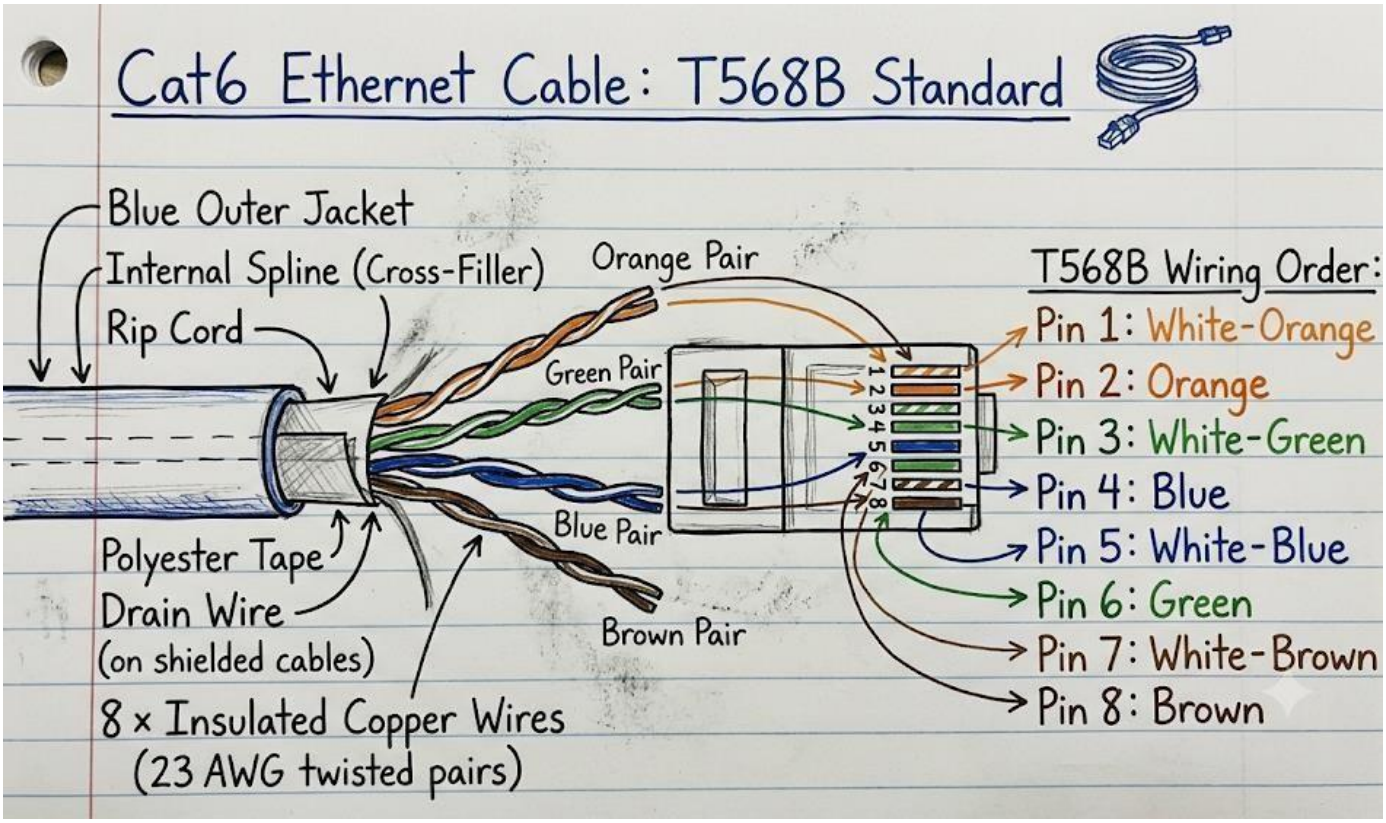
N+ - Network fundamentals and building Networks

Network cable and connectors

TASKS:1

Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

RESULT



Comptia N+

TASKS:-2

List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

RESULT

Network Cable	**Real-World App/Device**	**Why It Is Used**
Cat5e Ethernet Cable	Wi-Fi Router	It connects the router to a computer or network device and provides reliable wired network connectivity.
Cat6 Ethernet Cable	Zomato POS Terminal	It provides faster and more stable communication between the POS terminal and the local network.
Fiber Optic Cable	IPL Streaming Server	It supports very high-speed data transmission over long distances, which is useful for streaming large amounts of video data.

- Cat5e is commonly used for basic wired networking,
- Cat6 provides higher-speed Ethernet connections.
- fiber optic is suitable for high-speed, long-distance communication.

Comptia N+

TASKS:-3

Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

RESULT

These steps are following

1. **Cut the cable:** Cut the Ethernet cable to the required length using a cable cutter.
2. **Remove the outer jacket:** Carefully strip about 2–3 cm of the outer covering without damaging the internal wires.
3. **Untwist the wires:** Separate and straighten the eight small wires inside the cable.
4. **Arrange the wires:** Arrange the wires according to the **T568B standard** in this order:
   White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
5. **Trim the wires:** Make the eight wires equal in length and keep them straight.
6. **Insert into RJ-45 connector:** Insert the wires fully into the RJ-45 plug while keeping the correct color order.

7. **Crimp the connector:** Put the RJ-45 plug into the crimping tool and press it firmly so the metal contacts lock onto the wires.
8. **Check the connection:** Check that all wires are in the correct order and reach the end of the connector. If available, use a cable tester to verify the connection.

TASKS:-4

Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

RESULT

For an **IPL screening gaming zone**, I would use the following cables and connectors.

Comptia N+

Connection	**Cable**	**Connector**	**Reason**
Gaming console → TV/Monitor	HDMI 2.0/2.1	HDMI	Carries high-quality video and audio with low latency.
PC → Large display	**HDMI 2.1** or **DisplayPort**	HDMI/DisplayPort	Supports high resolution and high refresh rates for smooth gaming.
Multiple screens over long distance	Fiber-optic HDMI	HDMI	Useful for longer cable runs with less signal loss.
Console/PC → Network switch	Cat6 Ethernet	RJ-45	Provides fast, stable and low-latency network connectivity.
Network switch → Router	Cat6/Cat6a Ethernet	RJ-45	Reliable high-speed connection for multiple gaming devices.

Explanation:

I would mainly use **HDMI 2.1** for connecting consoles and gaming PCs to screens because it supports high-resolution video, high refresh rates and audio through a single cable. For network connections, **Cat6 Ethernet cables with RJ-45 connectors** are preferred because wired connections are more stable and have lower latency than Wi-Fi. For very long display connections, **fiber-optic HDMI** can help maintain signal quality.

Comptia N+

TASKS:-5

Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects. Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

RESULT



Photo: Ethernet cable connected to a Wi-Fi router.

Type of cable: Cat5e/Cat6 Ethernet cable
Connector: RJ-45 (8P8C) connector
Devices connected: Wi-Fi router and a computer/laptop or other Ethernet-enabled device.

Description:
The cable is plugged into the router's Ethernet/LAN port using an RJ-45 connector. It provides a **wired network connection** for transferring data between the router and the connected device. RJ-45 connectors are commonly used for Ethernet connections between routers, computers, switches, and other network devices.